



Implementation of a Big Data Pipeline for Transaction Anomaly Detection and Customer Behavior Analysis on an Online Retail Dataset Using Apache Spark, PostgreSQL, and MongoDB

Natan Obet Nego Hutahaean¹, Agatha C. S. Siahaan², Grace Advryanti Tampubolon³, Yos Ivan Sirait⁴, Samuel Rizki Sinambela⁵

Department of Software Engineering Technology, Faculty of Vocational, Institut Teknologi Del, Situluama, Indonesia

¹natanhutahaean01@gmail.com, ²agathasiahaan15@gmail.com, ³gracetampubolon94@gmail.com,

⁴ivansrt883@gmail.com, ⁵samuelsinambela0910@gmail.com

Abstract

The exponential growth of e-commerce transactions creates significant challenges in managing and analyzing massive data. This research aims to build an end-to-end Big Data pipeline to process large retail transaction datasets, maintain data quality, and detect anomalous transactions that indicate fraud or unusual behavior. The methods used include data extraction, transformation, and query (ETL) using Apache Spark and PySpark for distributed delivery. The dataset used is online retail transaction history sourced from Kaggle. The Isolation Forest Algorithm is implemented to detect anomalies in transaction data that has undergone a feature engineering process. The translation and anomaly detection results are stored in a dual-storage architecture: PostgreSQL as a relational data warehouse and MongoDB for storing anomaly documents. The results show that the pipeline successfully filtered the dirty data to 397,884 rows of clean, valid transactions. The Isolation Forest Model effectively identified 7,767 anomalous transactions. The final results are visualized through an interactive dashboard that presents business strategy insights such as revenue trends, geographic risk distribution, and best-selling product analysis. In conclusion, the implementation of this Big Data architecture can improve the efficiency of retail data collection and provide a strong foundation for data-driven business decision-making.

Keywords: transaction anomalies; Apache Spark; big data; isolation forest; customer behavior

How to Cite: [Caption completed by the editor]

Permalink/DOI: [Caption completed by the editor]

Received: [Caption completed by the editor]

Accepted: [Caption completed by the editor]

Available Online: [Caption completed by the editor]

This is an open-access article under the [CC BY 4.0 License](#)

Published by [Ikatan Ahli Informatika Indonesia](#)

1. Introduction

The development of information technology and the internet has driven the growth of e-commerce transactions exponentially [1]. Every day, online retail

platforms generate millions of transactions involving thousands of products and customers from all over the world [2]. The large volume of data, high production speeds, and diversity of data types create serious challenges in conventional data management and

analysis [3]. This condition is driving the adoption of Big Data technology as a solution to process and analyze data on a large scale [4]. One of the critical issues in the retail industry is the existence of anomalous transactions that can indicate system errors, fraud, or unnatural purchasing behavior [5]. Early detection of these kinds of anomalies is essential to maintain data integrity and protect the company's business losses [6].

In addition to anomaly detection, a deep understanding of customer behavior is also a key factor in modern business strategies [7]. Customer segmentation based on purchase patterns allows companies to design more targeted marketing strategies, improve customer retention, and optimize overall revenue [8]. Therefore, a comprehensive system is needed that is able to process large datasets, maintain data quality, and produce useful features to detect abnormal transactions [9].

This research aims to build an end-to-end Big Data pipeline that includes the process of ingest, cleaning, transformation, feature engineering, and storing retail transaction data. Furthermore, this study implements a transaction anomaly detection system using the Isolation Forest algorithm with Apache Spark and integrates PostgreSQL as a relational data warehouse and MongoDB as an anomaly-specific document store. The main contribution of this research is the provision of a scalable data processing architecture along with an interactive visual dashboard that clearly presents anomaly metrics and business insights to support strategic decision-making.

2. Methods

This study uses an applied quantitative approach by building an Extract, Transform, Load (ETL) based data pipeline architecture. The method stages are designed to handle the characteristics of Big Data which include volume, velocity, variety, veracity, and value [10]. Data processing utilizes Apache Spark and PySpark as the main engines due to their ability to efficiently perform distributed computing for large-scale data [11].

This study does not use a list of item or number formats, but describes the entire process in the form of descriptive paragraphs. This is done to comply with manuscript writing standards that prioritize sentence completeness and an integrated paragraph structure in describing each stage of the method.

2.1 Datasets and Data Pre-processing

The main data source used in this study is the online retail transaction history dataset from the UK obtained through the Kaggle platform [12]. The data includes variables such as invoice number, item code, product description, quantity, transaction time, unit price, customer identity, and country of origin [13].

The pre-processing stage (data cleaning) is focused on handling poor data quality. This process includes the

removal of blank rows specifically on customer identities, the filtering of illogical values, the removal of duplicate data, and the conversion of numeric data types. In addition, return transactions marked with negative-value quantities are identified and separated from normal sales transactions so as not to damage the revenue calculation [14].

2.2 Feature Engineering dan Anomaly Detection

After data cleansing, the feature engineering stage is carried out to extract new information that is essential to the analytical model [15]. The transaction time variable is broken down into hours, days, and months to identify customer purchase patterns [16]. The addition of features includes categorization of transaction revenue, purchase frequency, and average customer spend [17].

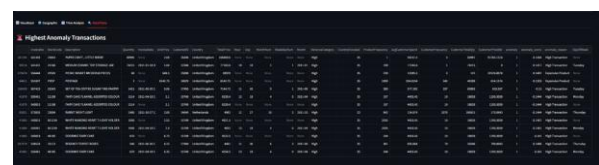
The enriched data is then used as input for the Isolation Forest model from the scikit-learn library [18]. This algorithm was chosen because of its ability to identify significantly different patterns (anomalies) than the majority of normal data [19]. The classification results provide binary labeling on each transaction, where a value of one represents an anomaly and a value of zero indicates a fair transaction [20].

2.3.Storage and Visualization Architecture

The data is temporarily stored in Apache Parquet format that supports lightweight compression and fast readout [21]. For final storage, the system architecture separates data into two types of databases to achieve polyglot persistence [22]. PostgreSQL is used as a relational database to store the entire net transaction data [23], while MongoDB is used as a document-based NoSQL database to store specific anomalous transaction history [24].

The final results are visualized using the Streamlit and Plotly libraries to build analytics dashboards that present performance metrics insights to end users [25]. This dashboard serves as the main interface that translates raw data into interactive graphical representations [26].

Table 1. Transaction anomaly data



Invoice Number	Item Code	Product Description	Quantity	Unit Price	Anomaly Count
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9
10	10	10	10	10	10

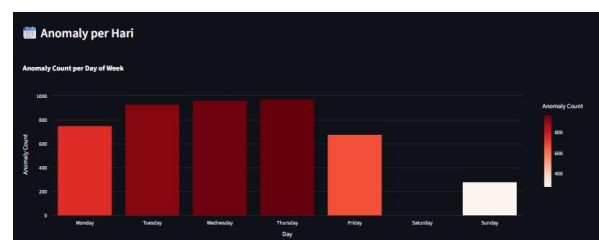


Figure 1. Anomaly Detection per day

3. Results and Discussions

This section presents the empirical evaluations of the Big Data pipeline execution alongside the strategic business insights extracted via the interactive Streamlit dashboard. The parallelized data cleaning routine executed over the Spark cluster exerted a profound impact on data quality, effectively insulating downstream machine learning models from garbage-in-garbage-out effects. The quantitative metrics contrasting the dataset state before and after the cleaning pipeline are explicitly detailed in Table 1.

Table 2. Quantitative Summary of the Data Cleaning Pipeline

Data Cleaning Report Metric	Pre-Cleaning State	Post-Cleaning State
Total Transaction Records	541,909 rows	397,884 rows
Minimum Item Quantity	-80,995 units	1 unit
Minimum Unit Price	-\$11,062.06	\$0.001
Missing Customer ID Rows	135,080 nulls	0 rows (Purged)
Dataset Retention Percentage	100.0%	73.4%

As structurally detailed in Table 1, the pipeline identified and removed 144,025 noisy or incomplete records. The primary driver behind this reduction was the absence of customer identification (CustomerID) affecting 135,080 rows, followed by the purge of 8,905 negative return items and 40 records with invalid unit prices. The refined final dataset retained 397,884 fully validated rows, presenting a pristine foundation for core predictive operations.

3.1 Processing Results and Anomaly Detection

The data cleansing process successfully filtered out tens of thousands of structural anomalies in the initial raw dataset. The pipeline execution results leave 397,884 lines of transaction data that are clean and valid. Through the feature engineering stage, the dataset structure is expanded to 21 informative attribute columns. Isolation Forest's machine learning model then processed the dataset and detected the presence of 7,767 transactions identified as anomalies.

This inference process increases the number of columns to 24 final attributes. The entire dataset, both normal and anomalous, is successfully stored in parallel into PostgreSQL and MongoDB databases without experiencing file size capacity constraints, demonstrating the scalability of the built architecture.

3.2 Business Insights and Customer Behavior Analysis

Based on the data that has been processed, analytics dashboards generate a number of measurable business insights. The system automatically calculates and

visualizes revenue trends daily and monthly net, as well as average order value. The geographic distribution analysis shows the distribution of the customer's country of origin while mapping the regions with the highest anomalous transaction ratio, which is very useful for the fraud analyst team. Product analysis identifies the ten items with the highest sales volume to support inventory management decisions.

In addition, the time feature extraction successfully maps the pattern of customer activity during peak hours through heatmap visualization, so that companies can optimally plan promotional campaign schedules and server capacity allocation. The return monitoring module independently evaluates the products with the highest cancellation rate as a material for future quality assurance review.

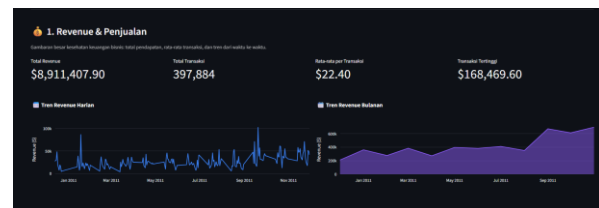


Figure 2. Dashboard

4. Conclusions

This research successfully implements an end-to-end Big Data pipeline that overcomes various challenges in massive-scale e-commerce data. The combination of Apache Spark for the ETL process and the Isolation Forest model proved effective, with the ability to filter the raw dataset down to 397,884 valid transactions and identify 7,767 suspicious transactions. The use of a hybrid storage architecture that combines PostgreSQL for structured data and MongoDB for anomalous documents provides high system flexibility. Additionally, the visual dashboard integration provides comprehensive monitoring tools regarding revenue metrics, risk areas, and product trends. These pipelines provide a solid operational foundation for extracting real business value from big data chunks in a structured and automated manner. Further development is expected to address the issue of finding synchronization latency between the database system and the dashboard application cache to provide real-time monitoring.

Acknowledgements

The author would like to thank the Software Engineering Technology Study Program, Vocational Faculty, Del Institute of Technology for the academic facilities and support provided during the implementation of this research project. The author states that there is no conflict of interest related to the funding or operation of this research.

References

- [1] N. Padhy, S. Suman, T. S. Priyadarshini, and S. Mallick, "A Recommendation System for E-

- Commerce Products Using Collaborative Filtering Approaches,” *Engineering Proceedings*, vol. 67, no. 1, p. 50, Sep. 2024, doi: 10.3390/engproc2024067050.
- [2] J.P. P. Khine and Z. Wang, “A Review of Polyglot Persistence in the Big Data World,” *Information*, vol. 10, no. 4, p. 141, Apr. 2019, doi: 10.3390/info10040141.
- [3] M. Alves Gomes and T. Meisen, “A review on customer segmentation methods for personalized customer targeting in e-commerce use cases,” *Information Systems and e-Business Management*, vol. 21, no. 3, pp. 527–570, Sep. 2023, doi: 10.1007/s10257-023-00640-4.
- [4] M. Mazhar, M. Siddique, H. Muqades, A. Hamza, and A. Khalid, “AI-BASED FRAUD DETECTION IN E-COMMERCE PAYMENTS: A COMPREHENSIVE RESEARCH STUDY,” *Contemporary Journal of Social Science Review*, vol. 4, no. 2, pp. 13–26, Apr. 2026, doi: 10.63878/cjssr.v4i2.2330.
- [5] P. M. Mah, I. Skalna, and T. Pelech-Pilichowski, “AI-Driven Anomaly Detection in E-Commerce Services: A Deep Learning and NLP Approach to the Isolation Forest Algorithm Trees,” *Journal of Theoretical and Applied Electronic Commerce Research*, vol. 20, no. 3, p. 214, Aug. 2025, doi: 10.3390/jtaer20030214.
- [6] U. Nayak, “AI-Powered Data Pipelines: Leveraging Machine Learning for ETL Optimization,” *Quest Journals Journal of Software Engineering and Simulation*, vol. 11, no. 6, pp. 134–136, Jun. 2025, doi: 10.35629/3795-1106134136.
- [7] A. Munshi, A. Alhindi, T. M. Qadah, and A. Alqurashi, “An Electronic Commerce Big Data Analytics Architecture and Platform,” *Applied Sciences*, vol. 13, no. 19, p. 10962, Oct. 2023, doi: 10.3390/app131910962.
- [8] E. Khashan, A. Eldesouky, and S. Elghamrawy, “An adaptive spark-based framework for querying large-scale NoSQL and relational databases,” *PLOS ONE*, vol. 16, no. 8, p. e0255562, Aug. 2021, doi: 10.1371/journal.pone.0255562.
- [9] S. Gupta and U. Sav, “Anomaly Detection in E-commerce Transactions Using Hybrid Deep Learning Models,” *Iconic Research And Engineering Journals*, vol. 9, no. 9, pp. 2392–2394, Mar. 2026, doi: 10.64388/IREV9I9-1715472.
- [10] J. Urnikienė, V. Steponavičienė, and S. Atanasov, “Comparative Read Performance Analysis of PostgreSQL and MongoDB in E-Commerce: An Empirical Study of Filtering and Analytical Queries,” *Big Data and Cognitive Computing*, vol. 10, no. 2, p. 66, Feb. 2026, doi: 10.3390/bdcc10020066.
- [11] T. Ozcan, “Customer Segmentation Using an Extended RFM Model and Clustering Algorithms in E-Commerce,” *Journal of Theoretical and Applied Electronic Commerce Research*, vol. 21, no. 5, p. 142, May 2026, doi: 10.3390/jtaer21050142.
- [12] Y. Weng and J. Wu, “Database management systems for artificial intelligence: Comparative analysis of postgres SQL and MongoDB,” *World Journal of Advanced Research and Reviews*, vol. 25, no. 2, pp. 2336–2342, Feb. 2025, doi: 10.30574/wjarr.2025.25.2.0586.
- [13] R. Singh, “Enhancing Financial Fraud Detection through Machine Learning: A Comparative Study of Anomaly Detection and Classification Models on Imbalanced Datasets,” *Proceedings of Botho University International Research Conference (BUIRC 2025)*, Atlantis Highlights in Sustainable Development, Dec. 2025, doi: 10.2991/978-94-6463-906-3_18.
- [14] I. Osiichuk and N. Zagorodna, “HYBRID DATA ARCHITECTURE IN E-COMMERCE,” *Advances in Cyber-Physical Systems*, vol. 11, no. 1, pp. 85–90, May 2026, doi: 10.23939/acps2026.01.085.
- [15] A. Sirimalla, “INTELLIGENT WORKLOAD PLACEMENT AND SHARDING BETWEEN ORACLE, POSTGRESQL, MONGODB, AND CASSANDRA USING MACHINE LEARNING,” *International Journal of Innovative Research in Engineering & Multidisciplinary Physical Sciences*, vol. 9, no. 3, May 2021, doi: 10.37082/IJIRMP.v9.i3.232851.
- [16] G. Pal, G. Li, and K. Atkinson, “Multi-Agent Big-Data Lambda Architecture Model for E-Commerce Analytics,” *Data*, vol. 3, no. 4, p. 58, Dec. 2018, doi: 10.3390/data3040058.
- [17] A. Hasnain, W. Arnaout, N. Faisal, A. B. B. Zainudden, A. N. Hamdan, and N. Amin, “Optimizing Retail Decision-Making Through Big Data and Machine Learning Integration,” *Preprints.org*, Dec. 2025, doi: 10.20944/preprints202512.0675.v1.
- [18] I. Waspada, N. Bahtiar, P. W. Wirawan, and B. D. A. Awan, “Performance Analysis of Isolation Forest Algorithm in Fraud Detection of Credit Card Transactions,” *Khazanah Informatika: Jurnal Ilmu Komputer dan Informatika*, vol. 6, no. 2, 2020, doi: 10.23917/khif.v6i2.10520.
- [19] U. Barman and K. Joshi, “Polyglot Persistence - Usage and Challenges,” *Journal of Artificial Intelligence and Machine Learning in Defense*, Nov. 2025, doi: 10.51219/JAIMLD/utpal-barma/633.
- [20] D. S. Jaybhay, A. B. Rasure, and R. R. Alapure, “REAL TIME BIG DATA ANALYTICS WITH APACHE SPARK,” *International Journal of Advanced Research in Computer and Communication Engineering*, vol. 14, no. 11, pp. 114–118, Nov. 2025, doi: 10.17148/IJARCC.2025.141118.
- [21] H. Carvalho and F. Maciel, “RFM-Based Customer Segmentation: A Pedagogical Case Study for Marketing Analytics Education,” *Journal for Advancement of Marketing Education*, vol. 34, no. 1, pp. 14–26, Feb. 2026, doi: 10.63963/001c.157562.
- [22] A. Karras, C. Karras, A. Pervanas, S. Sioutas, and C. Zaroliagis, “SQL Query Optimization in

Distributed NoSQL Databases for Cloud-Based Applications,” ALGO CLOUD 2022, LNCS 13799, pp. 21–41, 2023, doi: 10.1007/978-3-031-33437-5_2.

- [23] R. K. L. Kennedy, F. Villanustre, and T. M. Khoshgoftaar, “Unsupervised feature selection and class labeling for credit card fraud,” *Journal of Big Data*, vol. 12, no. 1, p. 111, Dec. 2025, doi: 10.1186/s40537-025-01154-1.
- [24] R. G. Martínez, R. A. Carrasco, C. Sanchez-Figueroa, and D. Gavilan, “An RFM Model Customizable to Product Catalogues and Marketing Criteria Using Fuzzy Linguistic Models: Case Study of a Retail Business,” *Mathematics*, vol. 9, no. 16, p. 1836, Aug. 2021, doi: 10.3390/math9161836.
- [25] R. G. Martínez, R. Carrasco, C. Sanchez-Figueroa, and D. Gavilan, “New fuzzy RFMD model. A method to identify new customer profiles due to an increase in online activity. Case study of a retail business,” *Studies of Applied Economics*, vol. 39, no. 8, Sep. 2021, doi: 10.25115/eea.v39i8.5524.
- [26] R. G. Martínez, R. A. Carrasco, J. García-Madariaga, C. P. Gallego, and E. Herrera-Viedma, “A comparison between Fuzzy Linguistic RFM Model and traditional RFM model applied to Campaign Management. Case study of retail business,” *Procedia Computer Science*, 2019, doi: 10.1016/j.procs.2019.11.286.